

Nicole Sandra-Yaffa Dumont

Ph.D. Candidate

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Education

Ph.D in Computer Science

Diploma in Theoretical Neuroscience

University of Waterloo

📅 Sept 2019 – March 2025

📍 Waterloo, Canada

- Dissertation title: *Symbols, Dynamics, and Maps: A Neurosymbolic Approach to Spatial Cognition*
- Research in computational neuroscience (spatial cognition & reinforcement learning) under Chris Eliasmith (Computational Neuroscience Research Group) and Jeff Orchard (NeuroCognitive Computing Lab)
- Neuro-Symbolic Modeling for Spatial Representation & Updating
 - Developed Spatial Semantic Pointers (SSPs) as a grid cell-based representation for continuous features. SSPs enable structured, hierarchical encoding across multiple domains (e.g., spatial, temporal, conceptual).
 - Built spiking neural network models for path integration and SLAM, leveraging SSPs for robust spatial inference on neuromorphic hardware.
- Reinforcement Learning & Action Selection
 - Designed biologically plausible online learning rules for reinforcement learning in continuous-time settings using State Space Models.
 - Applied SSPs & successor features for efficient navigation and policy learning.
 - Integrated SSPs into a Bayesian Optimization framework, improving active exploration in multi-agent environments.

Computational Mathematics (Masters of Mathematics – Co-op Program)

University of Waterloo

📅 Sept 2017 – April 2019

📍 Waterloo, Canada

- Completed a research project on robust optimization of an asset pricing model applied to carbon emissions, and course work on topics including optimization, computational statistics, numerical analysis, and deep learning.

Honours Mathematics and Physics (Bachelors of Science)

McMaster University

📅 Sept 2012 – April 2017

📍 Hamilton, Canada

Research

Publications & conferences

- **Dumont, N.S.**[†], Furlong, P.M.[†], Antonova, R.[†], Orchard, J., & Eliasmith, C. (submitted, in revision). Compositional neurosymbolic representations enable efficient active exploration. *Nature Communications*
- **Dumont, N.S.** (2025). *Symbols, dynamics, and maps: A neurosymbolic approach to spatial cognition* (PhD Thesis, University of Waterloo, Waterloo, ON) [\[Link\]](#)
- Furlong, P. M., Simone, K., **Dumont, N.S.**, Bartlett, M., Stewart, T. C., Orchard, J., & Eliasmith, C. (2024). Biologically-plausible markov chain monte carlo sampling from vector symbolic algebra-encoded distributions. In *International Conference on Artificial Neural Networks* (pp. 94–108). Springer [\[Link\]](#)
- Furlong, P. M., **Dumont, N.S.**, & Orchard, J. (2024). A recurrent dynamic model for efficient bayesian optimization. In *2024 Neuro Inspired Computational Elements Conference (NICE)* (pp. 1–5). IEEE [\[Link\]](#)
- **Dumont, N.S.**, Furlong, P. M., Orchard, J., & Eliasmith, C. (2023). Exploiting semantic information in a spiking neural slam system. *Frontiers in Neuroscience*, 17 [\[Link\]](#)
- **Dumont, N.S.**, Stöckel, A., Furlong, P. M., Bartlett, M., Eliasmith, C., & Stewart, T. C. (2023). Biologically-based computation: How neural details and dynamics are suited for implementing a variety of algorithms. *Brain Sciences*, 13(2), 245 [\[Link\]](#)
- **Dumont, N.S.**[†], Bartlett, M.[†], Simone, K.[†], Eliasmith, C., & Orchard, J. (3). Improving reinforcement learning with biologically motivated continuous state representations. In *International Conference on Cognitive Modelling (ICCM) 2023*, Amsterdam, Netherlands [\[Link\]](#)
- **Dumont, N.S.**, Orchard, J., & Eliasmith, C. (2022). A model of path integration that connects neural and symbolic representation. In *Proceedings of the Annual Meeting of the Cognitive Science Society* (Vol. 44, 44), Toronto, ON: Cognitive Science Society [\[Link\]](#)

- **Dumont, N.S.[†]**, Bartlett, M.[†], Furlong, M. P., & Stewart, T. C. (2022). Biologically-plausible memory for continuous time reinforcement learning. In *International Conference on Cognitive Modelling (ICCM) 2022*, Toronto, Canada [\[Link\]](#)
- Voelker, A. R., Blouw, P., Choo, X., **Dumont, N.S.**, Stewart, T. C., & Eliasmith, C. (2021). Simulating and predicting dynamical systems with spatial semantic pointers. *Neural Computation*, 33(8), 2033–2067 [\[Link\]](#)
- **Dumont, N.S.**, Stewart, T. C., & Eliasmith, C. (2021). Spiking neural network model of simultaneous localization and mapping with spatial semantic pointers. In *Computational and systems neuroscience (Cosyne) 2021*, Online [\[Link\]](#)
- Coleman, T. F., **Dumont, N.S.**, Li, W., Liu, W., & Rubtsov, A. (2021). Optimal pricing of climate risk. *Computational Economics*, 1–34 [\[Link\]](#)
- **Dumont, N.S.**, & Eliasmith, C. (2020). Accurate representation for spatial cognition using grid cells. In *Proceedings of the Annual Meeting of the Cognitive Science Society* (pp. 2367–2373). Toronto, Canada: Cognitive Science Society [\[Link\]](#)

[†] = These authors contributed equally

Teaching

- **CS 371: Introduction to Computational Mathematics (2022)**: I was the sole instructor for two sections of this course, teaching over 120 students. I prepared in-person lectures, video lectures, assignments, and tests covering a range of numerical methods.
- **CS 245: Logic and Computation (2019, 2020)**: I was a graduate Teaching Assistant and Instructional Apprentice for this introductory course on mathematical logic. I prepared and led tutorials, graded assignments, and assisted students during office hours.

Academic service

- **2025** – Co-organizer, Language and Thought Topic Area, Telluride Neuromorphic Cognition Engineering Workshop
- **2024** – Invited speaker, Language and Thought Topic Area, Telluride Neuromorphic Cognition Engineering Workshop
- **2024** – Instructor, Nengo Summer School
- **2023** – Instructor, Nengo Summer School
- **2025** – Reviewer, CogSci 2025
- **2024** – Reviewer, Frontiers in Neuroscience

Scholarships & Awards

- **2024** – European Neural Network Society (ENNS) Best Paper Award
- **2022** – Barbara Hayes-Roth Award for Women in Math and Computer Science
- **2020, 2019** – Go-Bell Scholarship
- **2019** – Provost Doctoral Entrance Award for Women
- **2019** – Keith & Debbie Geddes Graduate Scholarship
- **2014** – The Emanuel Williams Scholarship in Physics

Experience

Neuromorphic Scientist & Engineer

Applied Brain Research

 Jan 2023 – Jan 2024
 Waterloo, Canada

- Developed software to integrate a Spatial Semantic Pointer-based SLAM algorithm within a MuJoCo simulation environment and benchmarked it on neuromorphic hardware (Intel's Loihi 1 chip).

Research Associate

Cayuga Research

 May 2018 – Aug 2019
 Waterloo, Canada

- Consulting work focused on the development and implementation of optimization methods and data driven solutions to industrial problems.
- Projects included development of a global optimization toolbox in Matlab, prototype flight path optimization software, and chiller plant optimization & modelling.

Research Assistant

Ayers Research Group, McMaster University

 May 2015 – Aug 2015
 Hamilton, Canada

- Developed a semi-definite optimization algorithm to constrain a density matrix for representing a many-electron quantum system.